

Lab: Building and Running an Active Inference Agent

Learning Goal: Implement a complete agent, run it in an environment, and analyze its behavior.

Part 1: Complete the Agent

Exercise: Using the code from the module, create a foraging agent:

```
import numpy as np

# 3 locations: left, center, right
# Food is in the right location
# 3 observations: food, nothing, wall

A = np.array([
    [0.1, 0.1, 0.8],    # P(see food | state)
    [0.2, 0.8, 0.1],    # P(see nothing | state)
    [0.7, 0.1, 0.1]     # P(see wall | state)
])

B = np.zeros((3, 3, 3))
B[:, :, 0] = np.array([[0.9, 0.5, 0.0], [0.1, 0.4, 0.1], [0.0, 0.1, 0.9]])
B[:, :, 1] = np.array([[0.1, 0.1, 0.0], [0.8, 0.8, 0.2], [0.1, 0.1, 0.8]])
B[:, :, 2] = np.array([[0.0, 0.1, 0.1], [0.1, 0.4, 0.1], [0.9, 0.5, 0.8]])

C = np.array([3.0, 0.0, -1.0]) # prefer food, neutral to nothing, avoid wall
D = np.array([0.33, 0.34, 0.33]) # uniform start

# YOUR CODE: Create agent and environment, run simulation
```

Write your response here

Part 2: Analyze Behavior

Learning Goal: Interpret the agent's belief and action trajectories.

After running 20 steps:

1. Plot or list the agent's belief history — does the agent converge on the true state?
2. Plot or list the action history — does the agent find the food?
3. What does the true_state history show?

Write your response here

Part 3: Modify Agent Parameters

Learning Goal: Explore how parameters affect behavior.

Experiments:

1. Set $\gamma = 0.1$ (low policy precision) — run 20 steps. How does behavior change?
2. Set $\gamma = 10.0$ (high policy precision) — run 20 steps. How does behavior change?
3. Set $C = [0, 0, 0]$ (no preferences) — what does the agent do now?
4. Make A very noisy (nearly uniform) — how does this affect inference?

Write your response here

Part 4: Add Logging

Learning Goal: Implement better observation and analysis.

Exercise: Add a method to the agent class that prints a summary:

```
def summary(self):  
    """Print a summary of the agent's experience."""  
    print(f"Steps taken: {len(self.action_history)}")  
    print(f"Action distribution: {np.bincount(self.action_history, minlength=10)}")  
    print(f"Final beliefs: {self.qs.round(3)}")  
    print(f"Observation counts: {np.bincount(self.obs_history, minlength=10)}")
```

Run the agent for 50 steps and call `summary()`. What do you observe?

Write your response here

Part 5: Reflection

In 150 words, reflect: What surprised you about the agent's behavior? Did it always find the food? When did it explore vs. exploit? How does watching the code run deepen your understanding compared to the mathematical description?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Full implementation	Agent + Environment + Loop
2	Analysis	Belief and action trajectories
3	Parameter exploration	gamma, C, A effects
4	Logging	Diagnostic reporting
5	Reflection	Code ↔ Theory connection